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GEOG 411
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Unit 4: Image Classification

I ran into a lot of hiccups when trying to execute both types of image classification.

For Object-based classification, the Wizard typically got stuck at 12.5 - 25% in the 'Train' window:

- "Generating Segmented Dataset..."; "Generate Raster from Raster Function ..."
- "Distributing operation across 6 parallel instances."

One attempt ran for over 4 hours. Later, after restarting ArcGIS Pro, it was able to generate a preview of the classified results. But next, the wizard got stuck at 50% when exporting the dataset and its class definition file in the 'Classify' window:

- "Generating Classified Dataset..."; "Generate Raster from Raster Function ..."
- "Distributing operation across 16 parallel instances."

For Pixel-based classification, the Wizard was able to generate a preview of the classified results, but similarly got stuck at 50% when exporting the dataset and its class definition file in the "Classify" window:

- "Generating Classified Dataset..."; "Generate Raster from Raster Function ..."
- "Distributing operation across 16 parallel instances."

My longest attempt ran for 40 minutes. However, after canceling the process I found a workaround: I could simply export the preview raster results, outside the Wizard.

My ArcGIS project's File Geodatabase now contains four versions of results:

- 'pixelbased_classified' - raster whose Wizard export was canceled; evident in its reduced extent
- 'pixelbased_classified_expprev' - exported raster of the preview raster created by the Wizard
- 'objectbased_classified' - raster whose Wizard export was canceled; evident in its reduced extent
- 'objectbased_classified_expprev' - exported raster of the preview raster created by the Wizard

*I will present all four results, but I will only perform Accuracy Assessment for the full images—those which I exported from the Wizard's preview rasters ('*_expprev').*

Note that my segmented image was also created by exporting the preview of the segmented image that the Image Classification Wizard created after filling out the parameters in the 'Segmentation' window.

I regret not considering this earlier, but I think that one factor in my difficulty may have been that I keep all of my work on an external SSD. Perhaps if all of my data was saved and being written to my computer's C:\ drive, I would not run into such slow speeds/complete stops in the Wizard processes.

1. Below is the segmentation parameter space I used for Object-based classification:

- 'Spectral detail' = 14
- 'Spatial detail' = 16
- 'Minimum segment size in pixels' = 4

2. I chose these values for my parameter space because:
 - A greater spectral detail (14) can better discern between similar-looking 'developed' and 'barren' pixels; and 'ridge' and 'barren' pixels
 - A greater spatial detail (16) can better classify smaller features, like thin plots of 'planted' land cover
 - A small segment size (4) because **segments of (4 pixels * 30m/pixel) = 120m** ensures sensitivity to thinner features like roads, runways, lawns, or water
3. I created my classification schema by adapting and simplifying the NLCD2011.

classname	classvalue
water	0
developed	1
barren	2
ridge	3
planted	4

4. I kept designations from NLCD2011 like 'water', 'developed', 'barren', and 'planted' because they accounted for most land cover types in our image of Greater Las Vegas. While NLCD's designation for 'planted' land cover typically refers to pasture and cropland, I kept the name here because I thought it was suitable for the vegetation in urban Las Vegas like at golf courses, which appears prominently in a CIR composite. I eliminated NLCD's 'forest' classes altogether, because there are virtually no instances of forest land cover in Greater Las Vegas. I did the same for 'wetlands' and 'shrubland', and created a fifth class called 'ridge' to classify features distinct from 'barren' land due to their topographic or geologic complexity.

5.

classname	classvalue	#/samples	pixel%
water	0	10	14.31
developed	1	18	7.90
barren	2	9	50.10
ridge	3	16	26.58
planted	4	20	1.11

See Map 3; Map 4

6.

Object-based classification results: Confusion Matrix

	C_0	C_1	C_2	C_3	C_4	Total	U_Acc	Kappa
C_0	9	0	1	0	0	10	0.9	0
C_1	0	7	3	0	0	10	0.7	0
C_2	0	0	5	5	0	10	0.5	0
C_3	0	1	11	26	0	38	0.684211	0
C_4	0	6	1	1	2	10	0.2	0
Total	9	14	21	32	2	78	0	0
P_Acc	1	0.5	0.238095	0.8125	1	0	0.628205	0
Kappa	0	0	0	0	0	0	0	0.486842

Pixel-based classification results: Confusion Matrix

	C_0	C_1	C_2	C_3	C_4	Total	U_Acc	Kappa
C_0	5	0	5	0	0	10	0.5	0
C_1	0	8	1	1	0	10	0.8	0
C_2	0	0	10	3	0	13	0.769231	0
C_3	0	0	10	19	0	29	0.655172	0
C_4	0	4	2	3	1	10	0.1	0
Total	5	12	28	26	1	72	0	0
P_Acc	1	0.667	0.357143	0.730769	1	0	0.597222	0
Kappa	0	0	0	0	0	0	0	0.462687

6, cont.

User's accuracy: 1 - Commission Error Rate

class	Object-based	Pixel-based
water	90%	50%
developed	70%	80%
barren	50%	76.9231%
ridge	68.4211%	65.5172%
planted	20%	10%

Producer's accuracy: 1 - Omission Error Rate

class	Object-based	Pixel-based
water	100%	100%
developed	50%	66.6667%
barren	23.8095%	35.7143%
ridge	81.25%	73.0769%
planted	100%	100%

Overall accuracy:

Object-based classification: 62.8205% > Pixel-based classification: 59.7222%

Object-based Kappa: 0.486842 > Pixel-based Kappa: 0.462687

Both classification methods struggled to distinguish between the 'developed' and 'barren' classes, which were quite similar spectrally. These are also the only two classes where Pixel-based classification outscored Object-based classification.

I am surprised that both classification methods struggled to identify the 'planted' class, which was spectrally obvious, due to the IR color composite. While the 'planted' / or green land features were relatively small, they were great in number.

6, cont.

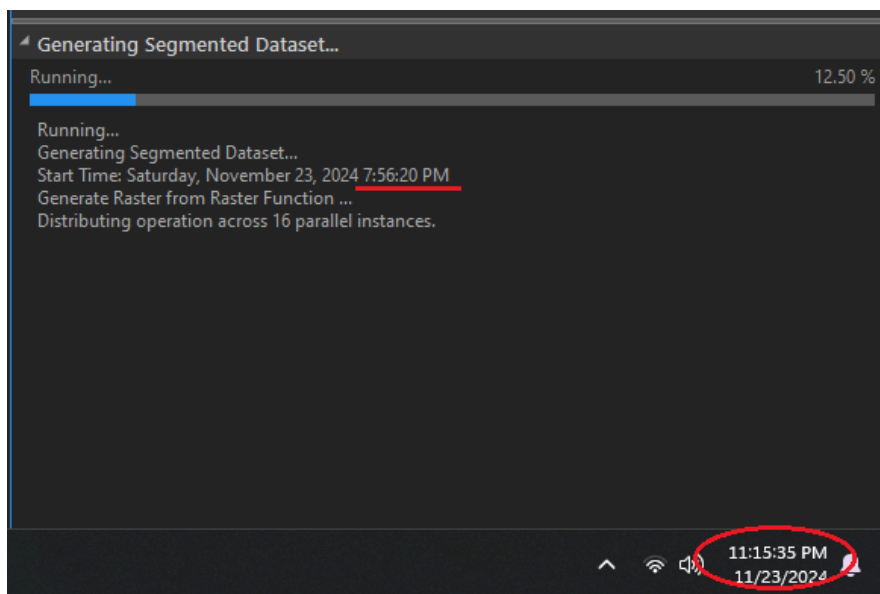
Although Object-based classification performed slightly better, neither method was very accurate. I suspect this is due to two glaring issues:

- **Poor class definition:** The confusion matrices prove that classes 'developed' and 'barren' are too spectrally similar to be accurately sorted. Before obtaining results, I thought that classes 'ridge' and 'barren' would prove confusing too. I've struggled to improve my classification schema.
- **Too few training samples:** I used only 9-20 training samples for each of my classes. I ran through the 'Train' window of the Image Classification Wizard many times, and I generated a greater number of training samples on my earlier attempts, when I had more patience. But I did not save those training datasets, and only have the one depicted in Map 3, Map 4, and described in table in #5 above. I used this small set of training samples because it is the one that *worked*, but my models would have greatly benefited from more samples.

7. See Map 8b

8. See Map 7b

Below: An unsuccessful attempt of Object-based classification at t = 3 hr 19 min



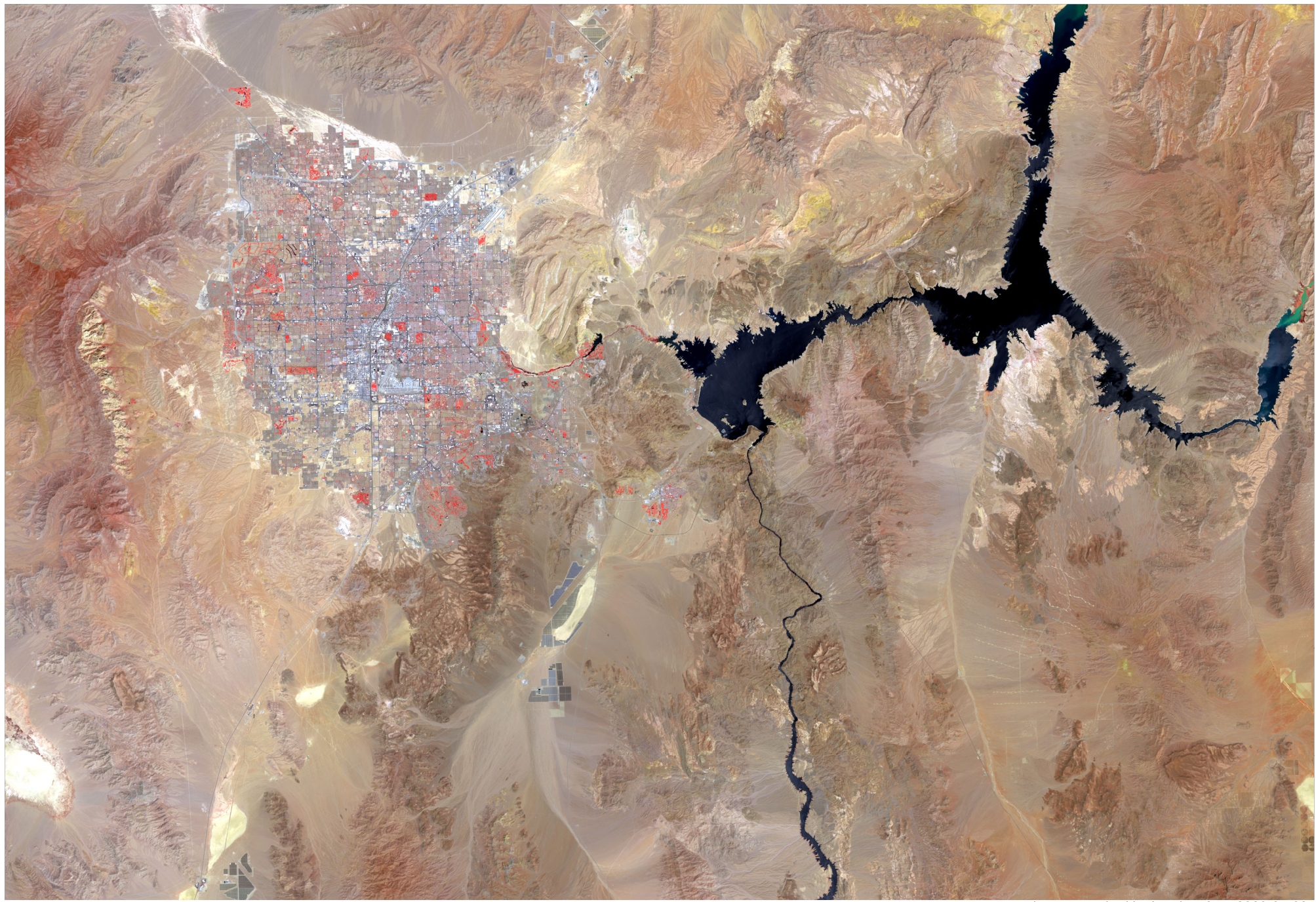


image acquired by Landsat 8 on 2020-05-02

Map 1: Greater Las Vegas (Landuse Composite)

WGS 1984 UTM Zone 11N

X: 114°50'53"W 0 4.25 8.5 17 Miles

Y: 35°59'47"N

Scale: 1:520,000

■ NIR (B5)
■ RED (B4)
■ BLUE (B2)

cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data

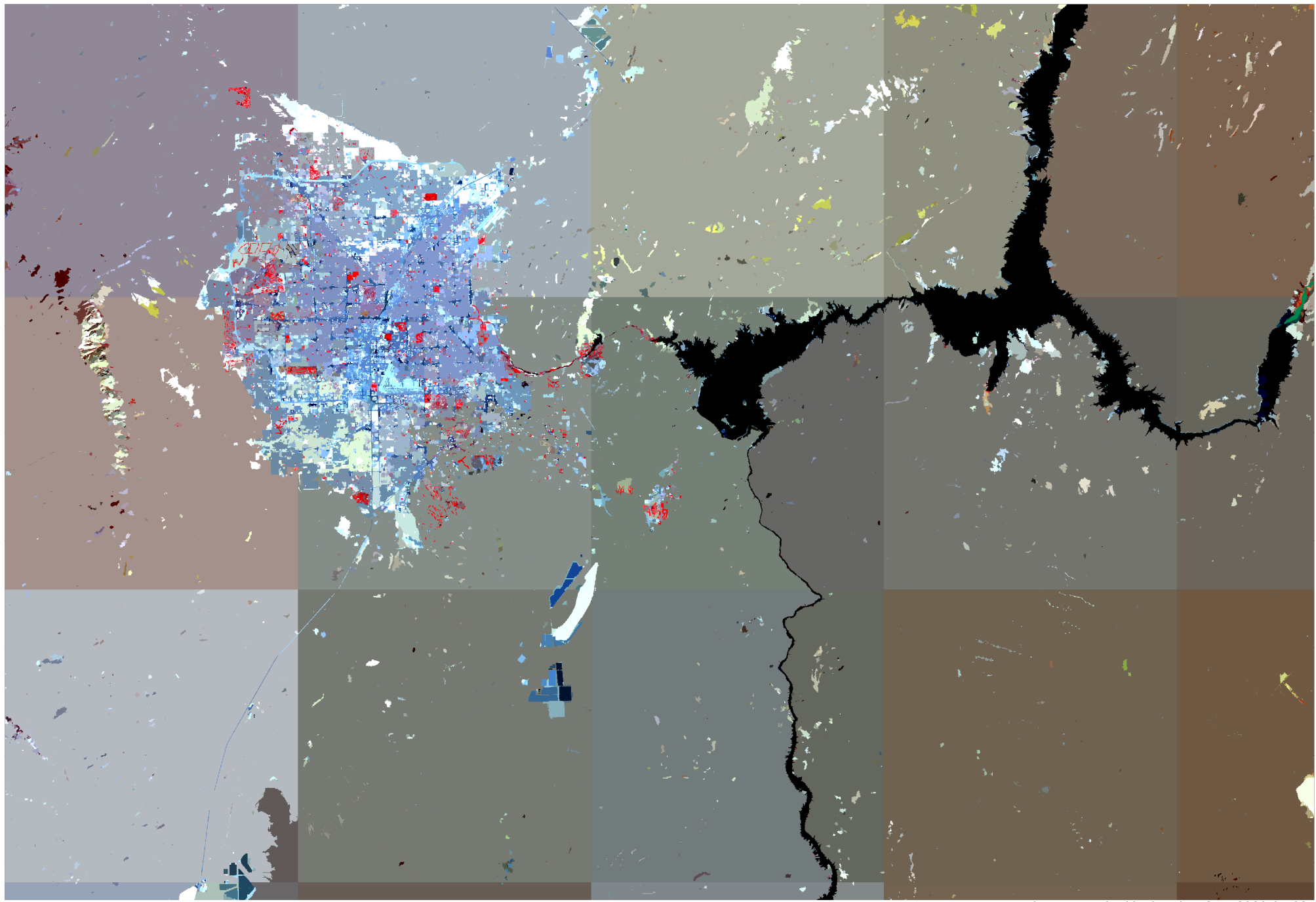


image acquired by Landsat 8 on 2020-05-02

Map 2: Segmented Image

WGS 1984 UTM Zone 11N

X: 114°50'53"W 0 4.25 8.5 17 Miles

Y: 35°59'47"N

Scale: 1:520,000

cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data

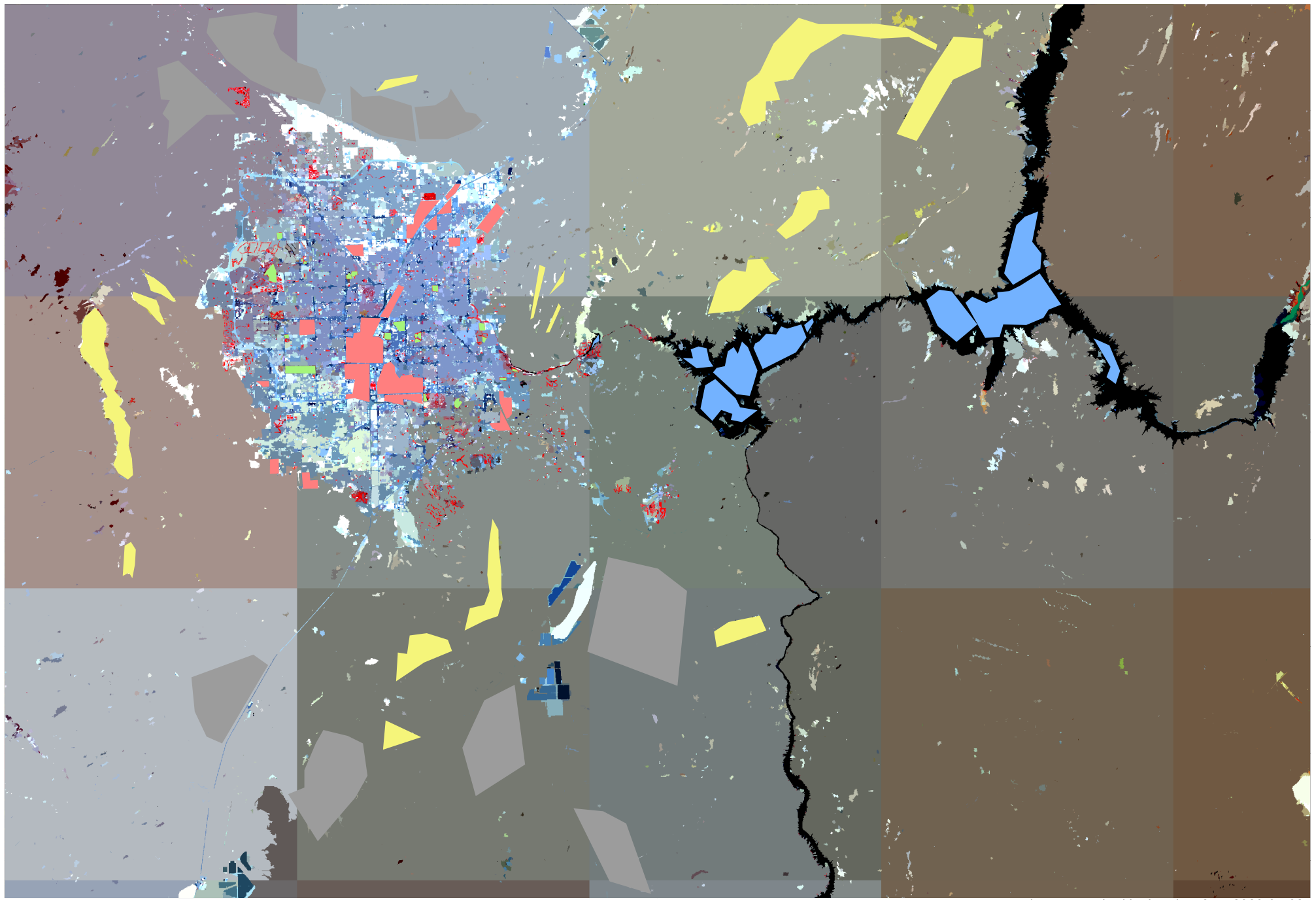


image acquired by Landsat 8 on 2020-05-02

Map 3: Training Samples over Segmented Image

WGS 1984 UTM Zone 11N

X: 114°50'53"W 0 4.25 8.5 17 Miles

Y: 35°59'47"N

Scale: 1:520,000

water
developed
barren

ridge
planted

cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data

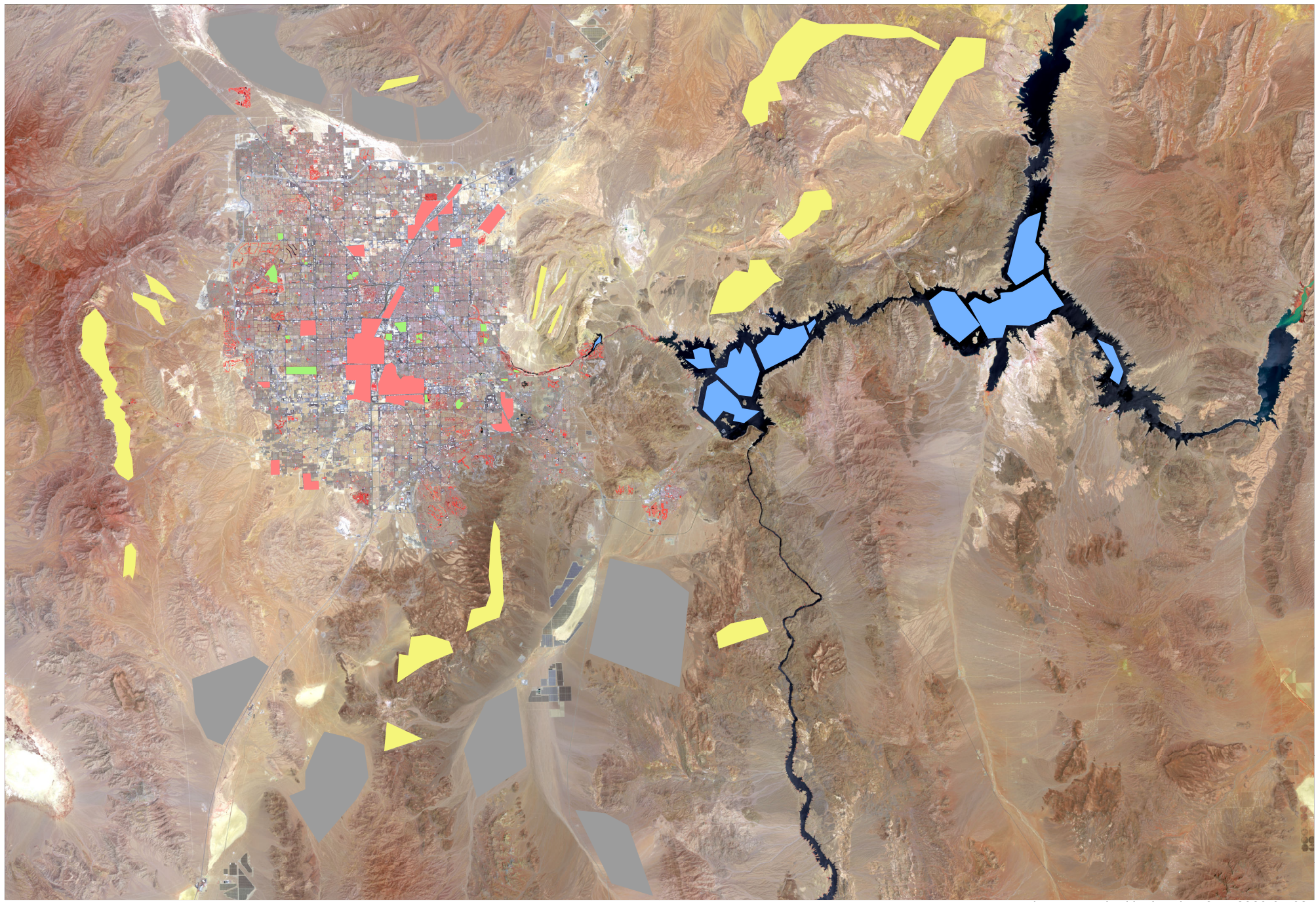


image acquired by Landsat 8 on 2020-05-02

Map 4: Training Samples over Source Image

WGS 1984 UTM Zone 11N

X: 114°50'53"W 0 4.25 8.5 17 Miles

Y: 35°59'47"N

Scale: 1:520,000



cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data

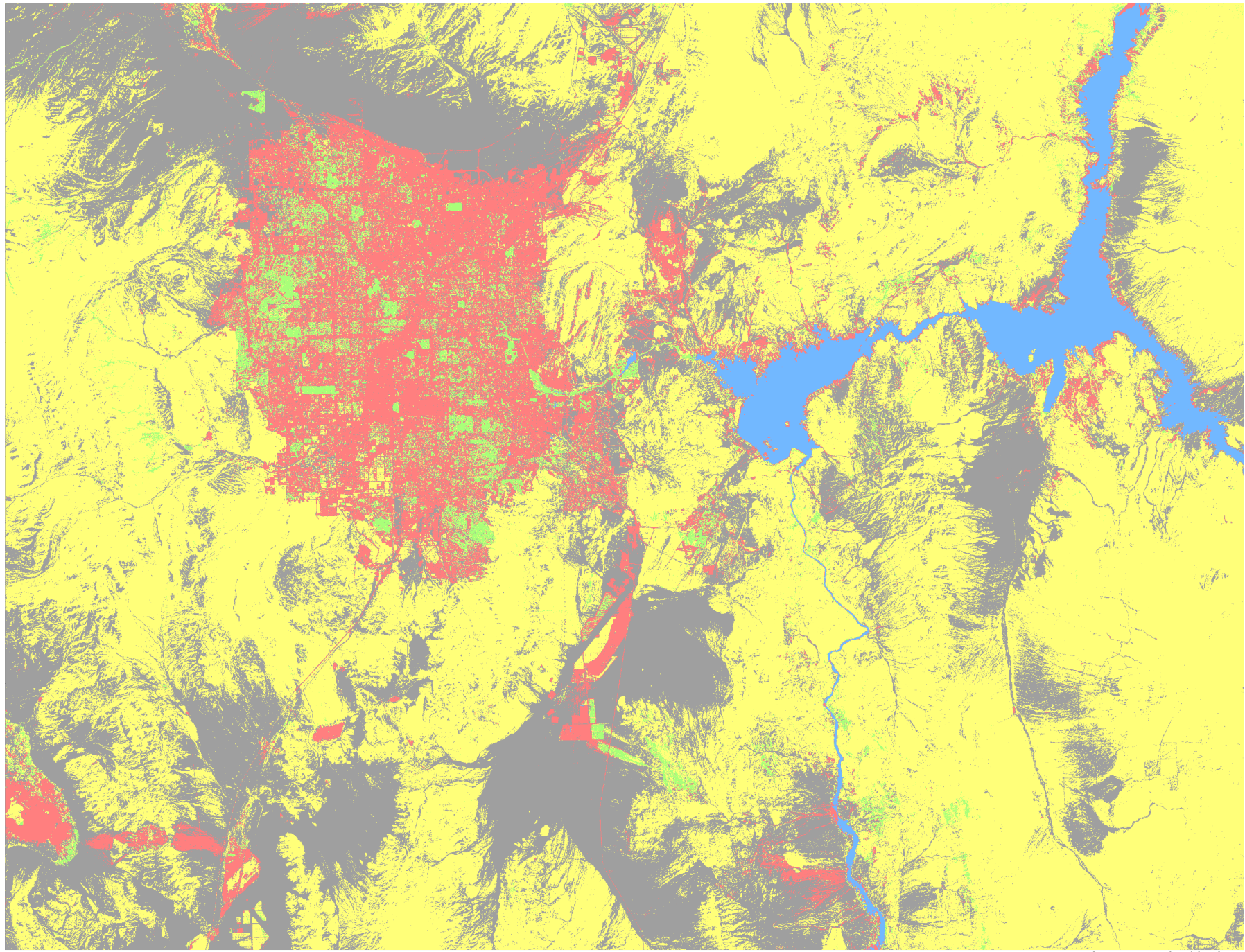


image acquired by Landsat 8 on 2020-05-02

Map 5: Pixel-based classification results (failed export)

WGS 1984 UTM Zone 11N
X: 114°50'53"W
Y: 35°59'47"N
Scale: 1:520,000

- water
- developed
- barren
- ridge
- planted

cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data

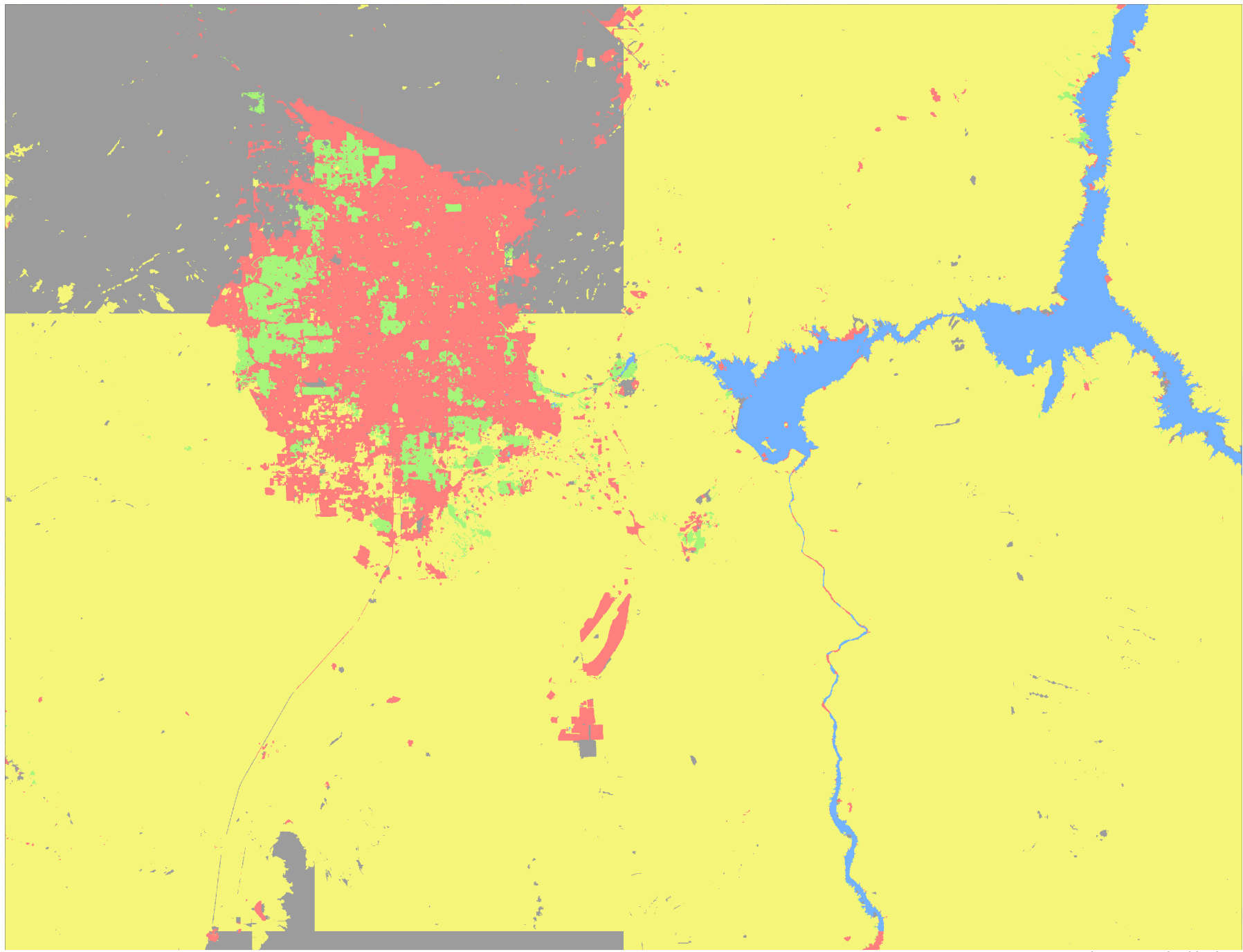


image acquired by Landsat 8 on 2020-05-02

Map 6: Object-based classification results (failed export)

WGS 1984 UTM Zone 11N
X: 114°50'53"W
Y: 35°59'47"N
Scale: 1:520,000

- water
- developed
- barren
- ridge
- planted

cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data

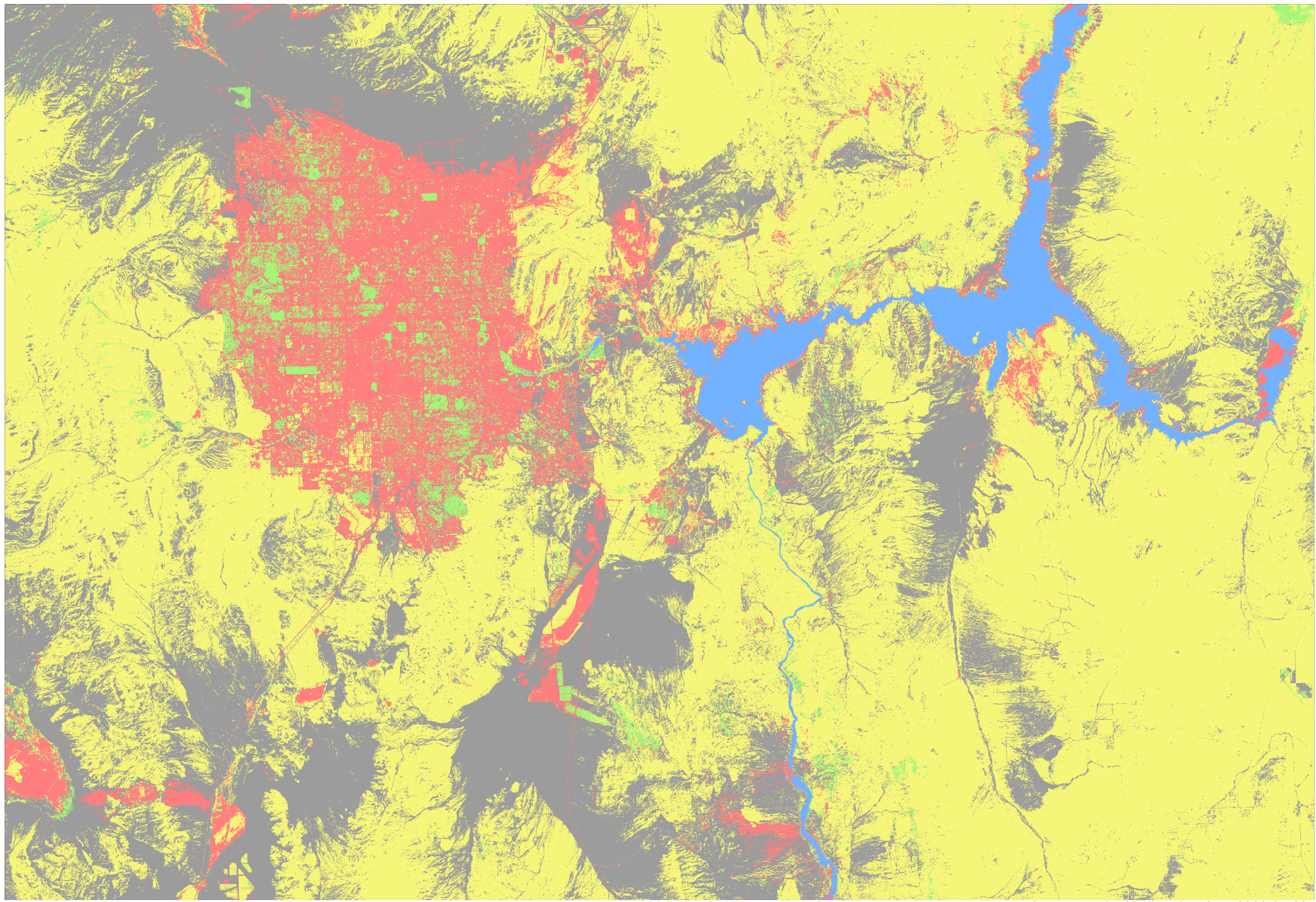


image acquired by Landsat 8 on 2020-05-02

Map 7a: Pixel-based classification results

WGS 1984 UTM Zone 11N

X: 114°50'53"W 0 4.25 8.5 17 Miles

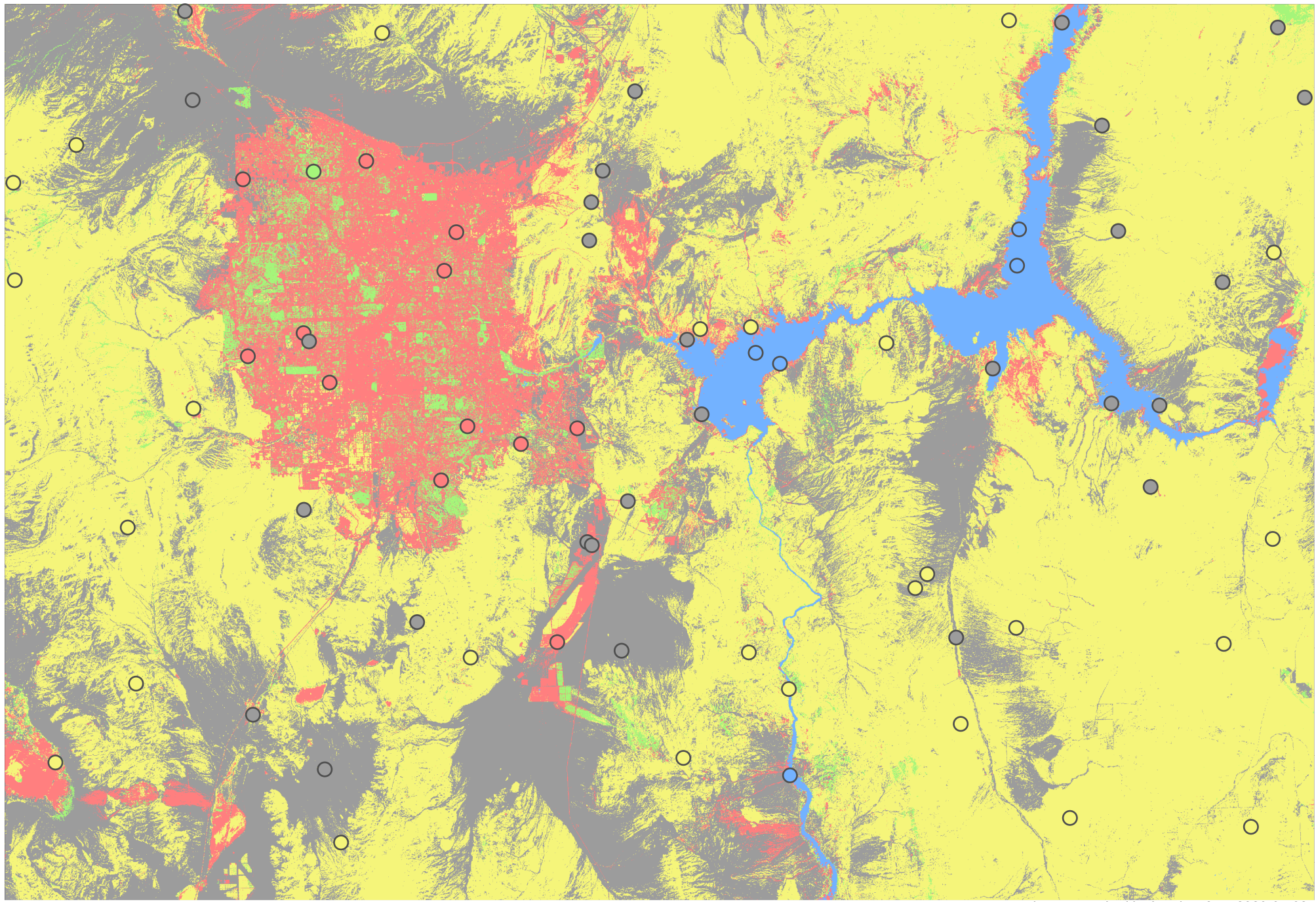
Y: 35°59'47"N

Scale: 1:520,000

water
developed
barren

ridge
planted

cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data



Map 7b: Pixel-based classification results w/ 72 A.A.P.

image acquired by Landsat 8 on 2020-05-02

WGS 1984 UTM Zone 11N
X: 114°50'53"W
Y: 35°59'47"N
Scale: 1:520,000

- water
- developed
- barren
- ridge
- planted

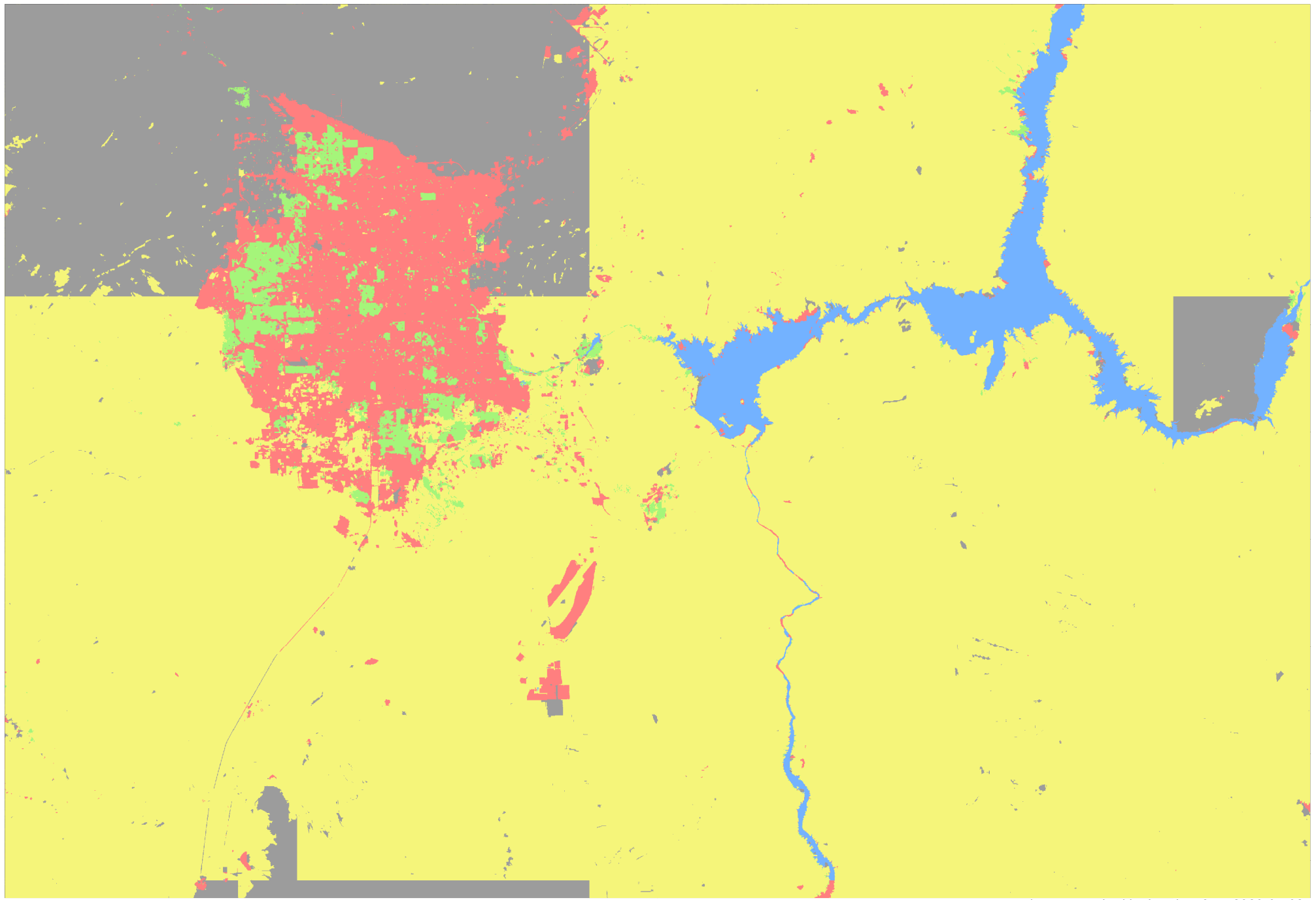


image acquired by Landsat 8 on 2020-05-02

Map 8a: Object-based classification results

WGS 1984 UTM Zone 11N

X: 114°50'53\"W 0 4.25 8.5 17 Miles

Y: 35°59'47\"N

Scale: 1:520,000

water
developed
barren

ridge
planted

cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data

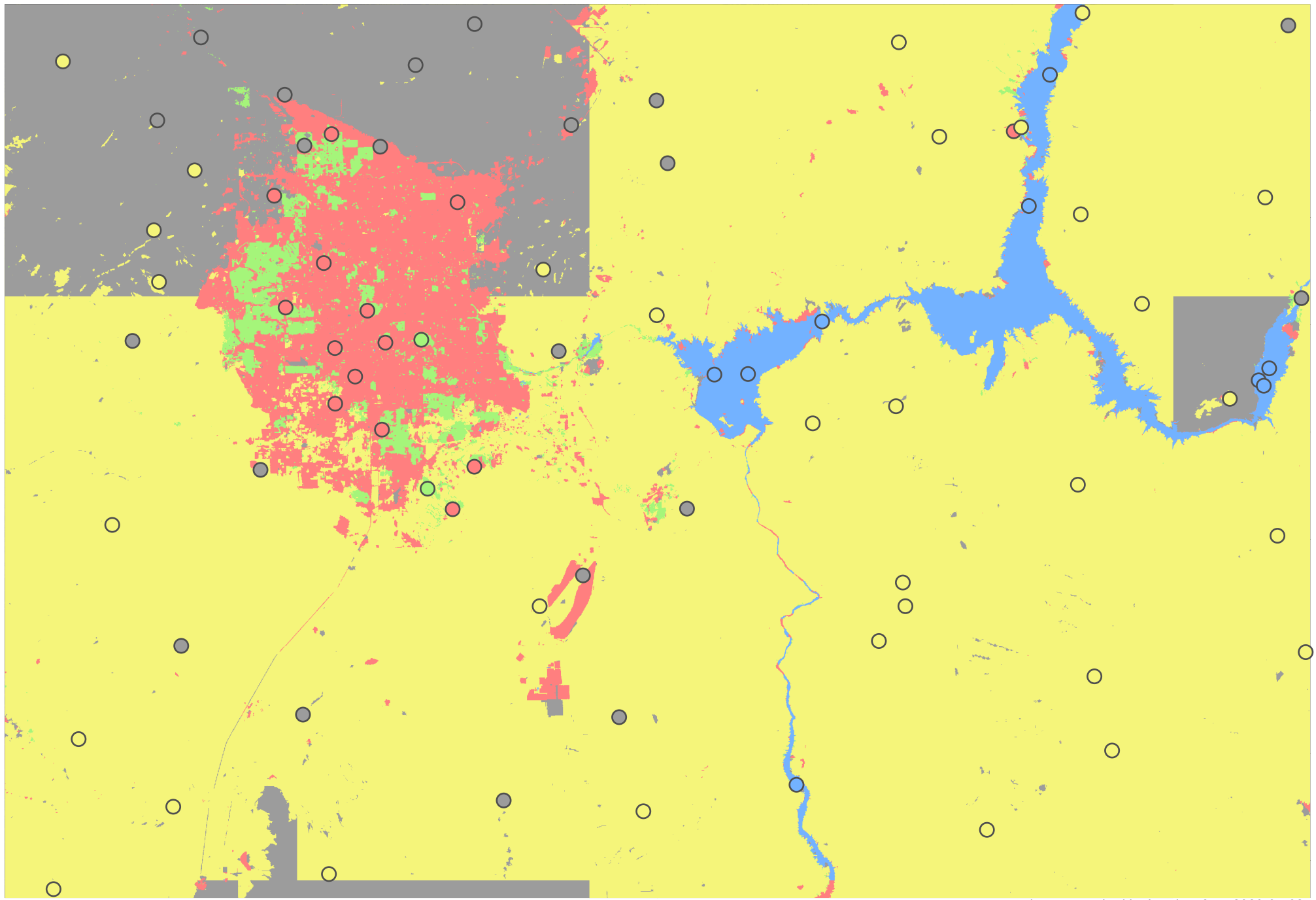


image acquired by Landsat 8 on 2020-05-02

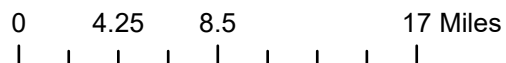
Map 8b: Object-based classification results w/ 78 A.A.P.

WGS 1984 UTM Zone 11N

X: 114°50'53\"/>

Y: 35°59'47\"/>

Scale: 1:520,000



cartography by Aaron Goodman for UCLA MAGIST GEOG 411 Fall 2024
data source: Landsat 8-9 Operation Land Imager and Thermal Infrared Sensor Collection 2 Level-1 Data